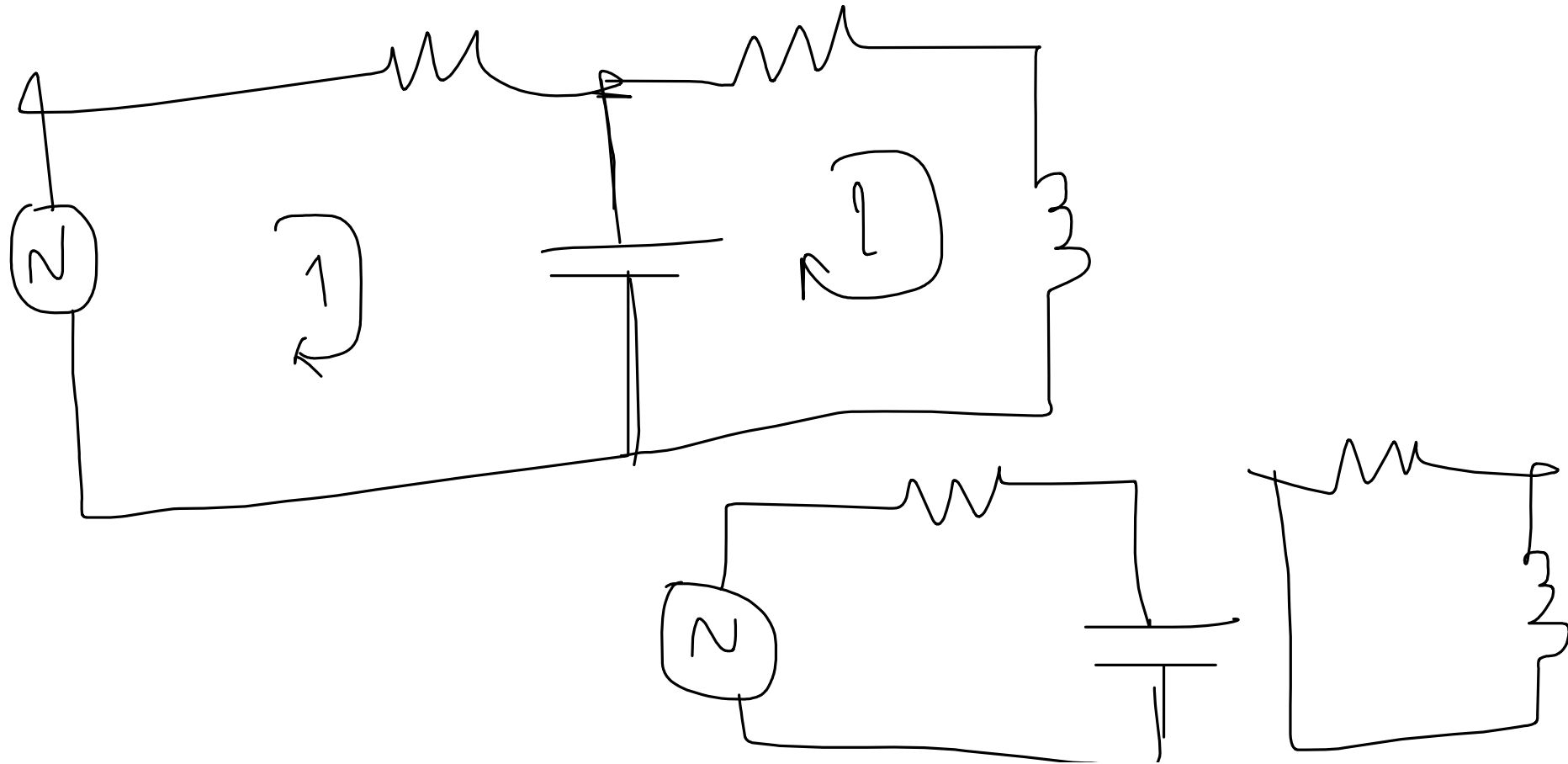
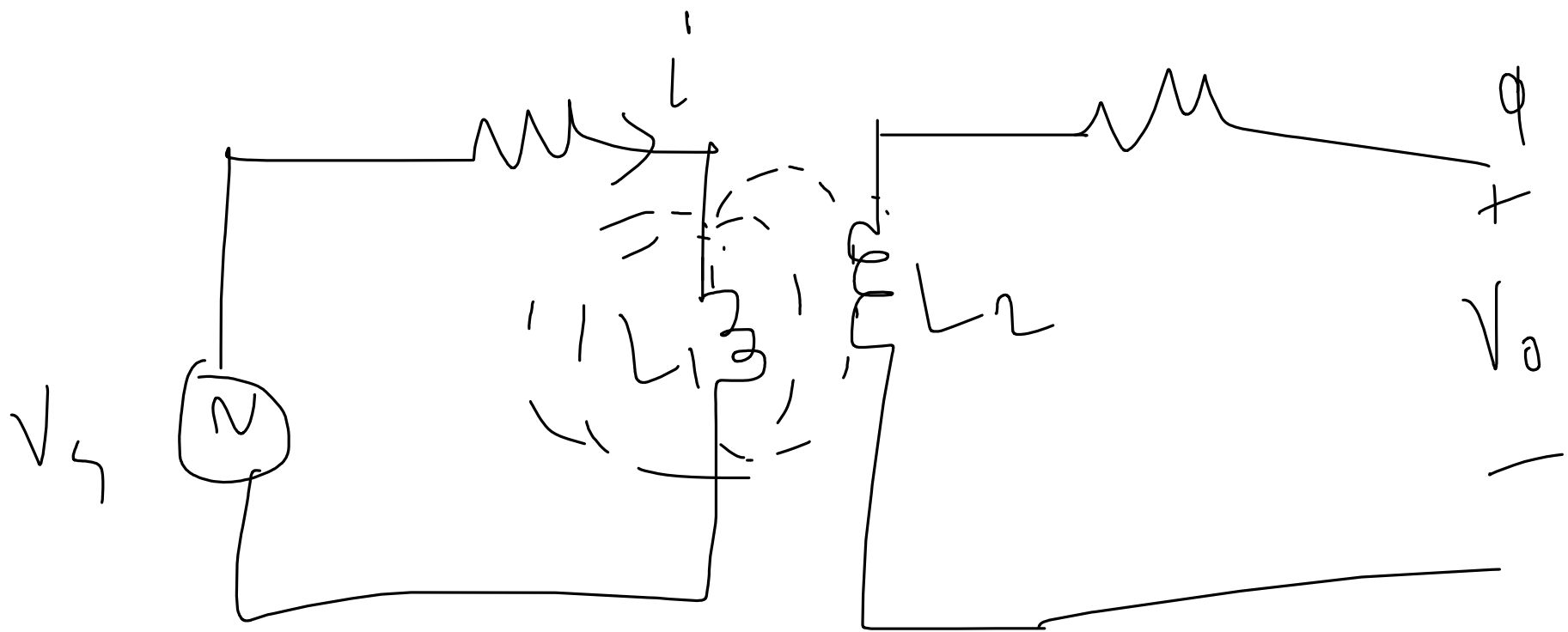


Magnetically Coupled ckt





ckt-2

magnetically coupled ckt



2 ckt's linked via magnetic
flux.

Condⁿ → 2 inductors must be close
enough to each other



$$V = N \times \frac{d\phi}{dt} \quad \checkmark$$

turn of coil

change of flux
wrt time

$i(t)$

DC $\rightarrow \frac{d\phi}{dt} \rightarrow 0$

$i(t) \rightarrow AC$

$\hookrightarrow$

θ

$\rightarrow$

changing

$\frac{d\theta}{dt}$

$\rightarrow$

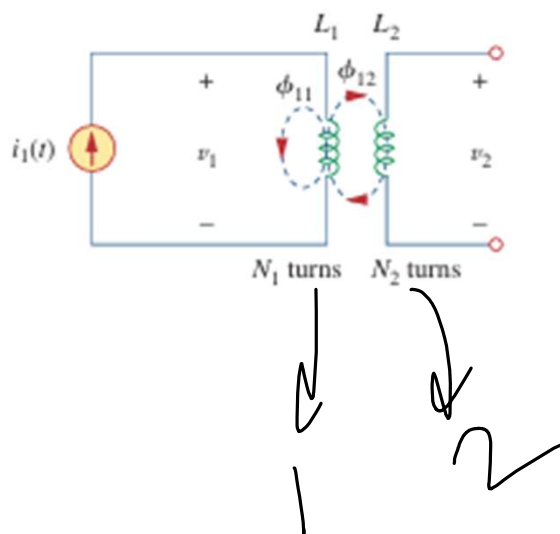
extnt

$$V = N \frac{dQ}{dt}$$

$$= \left[N \times \frac{dQ}{di} \right] \times$$

$$\frac{di}{dt} \checkmark$$

$$\boxed{V = L \frac{di}{dt}} \checkmark \checkmark$$



$$\Phi_1 = \Phi_{11} + \Phi_{12}$$

Voltage induced in coil-1

$$-V_1 = N_1 \times \frac{d\Phi_1}{dt}$$

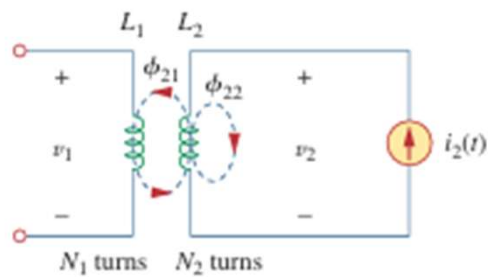
Voltage induced in coil-2

$$-V_2 = N_2 \times \frac{d\Phi_{12}}{dt}$$

$$V_1 = L_1 \frac{di_1}{dt}$$

$$V_2 = M_{21} \frac{di_1}{dt}$$

Mutual
inductance.

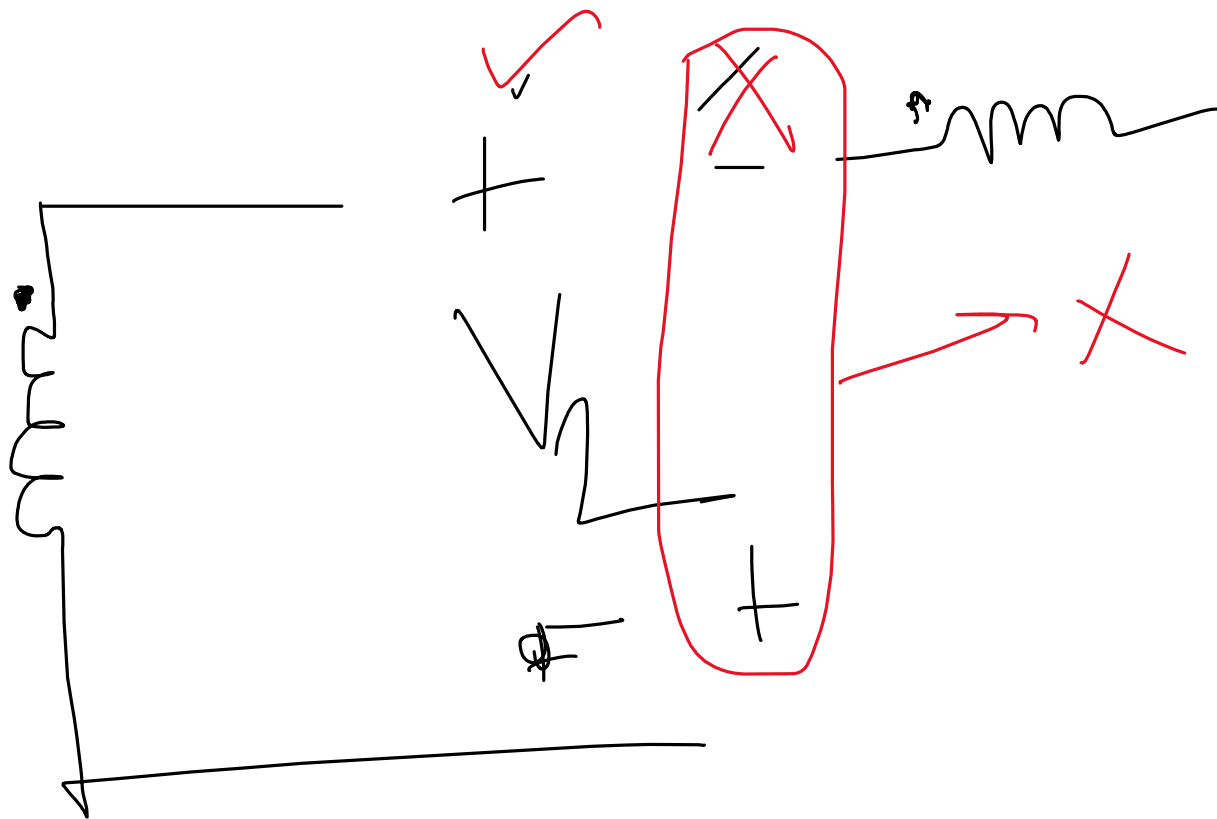
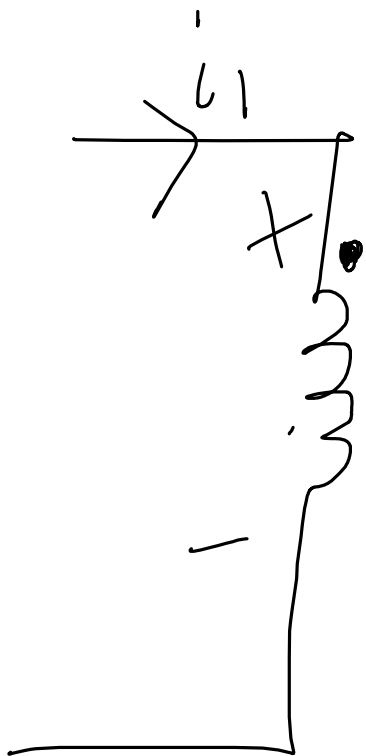


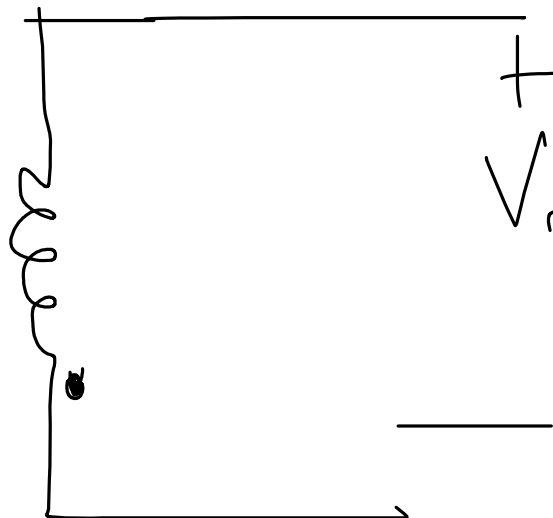
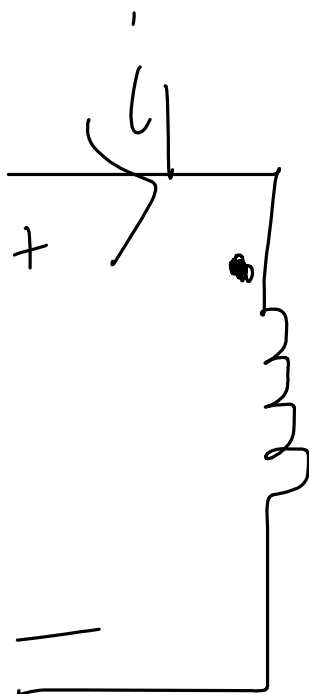
$$v_1 = M_{12} \frac{di_2}{dt}$$

Dot Convention

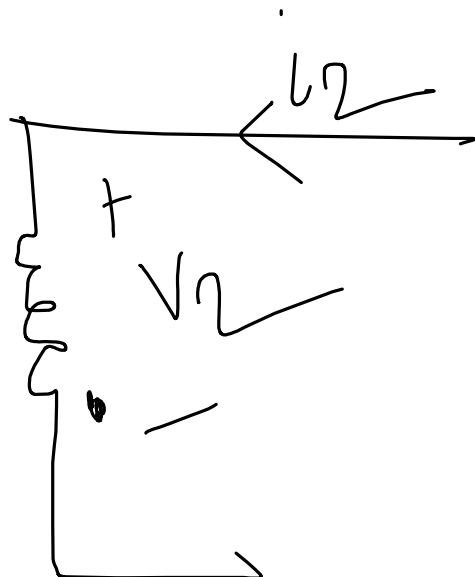


the way or technique to find
the polarity of induced voltage

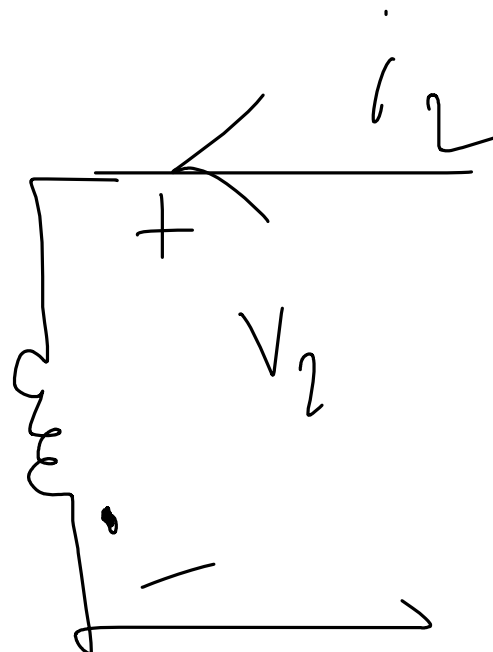


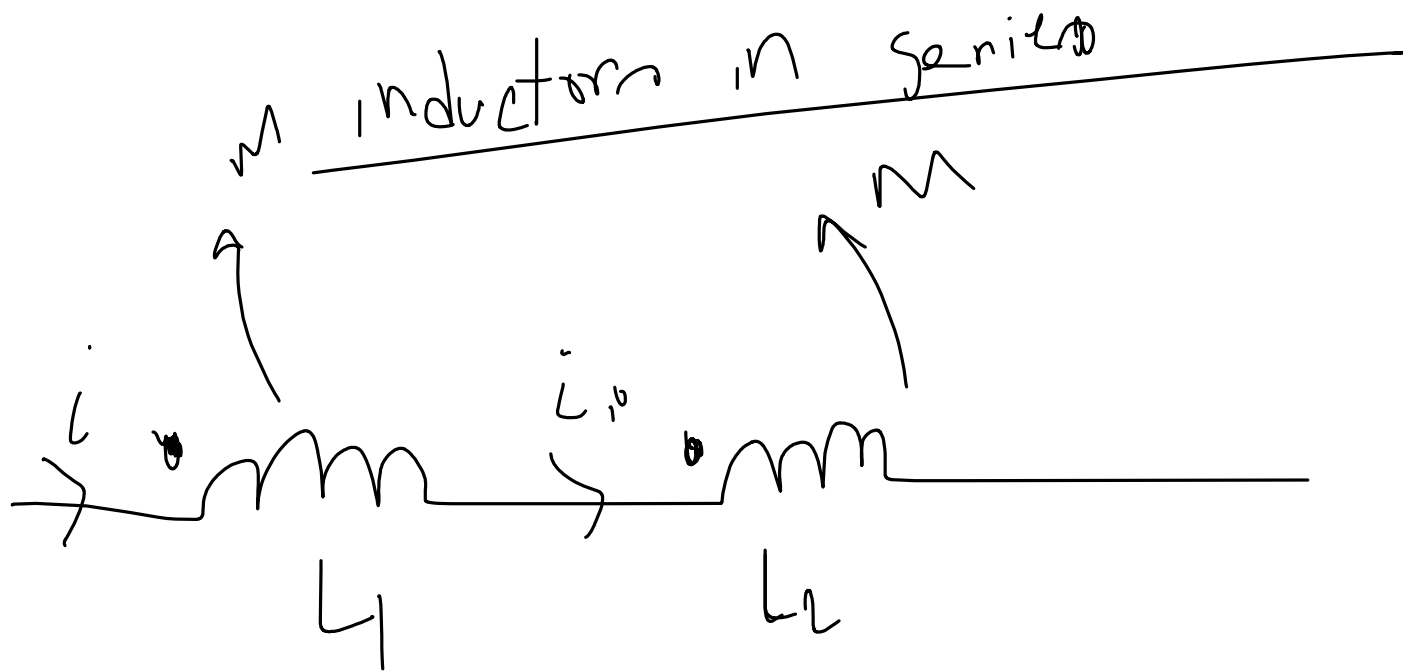


$$V_2 = M \frac{di_1}{dt}$$

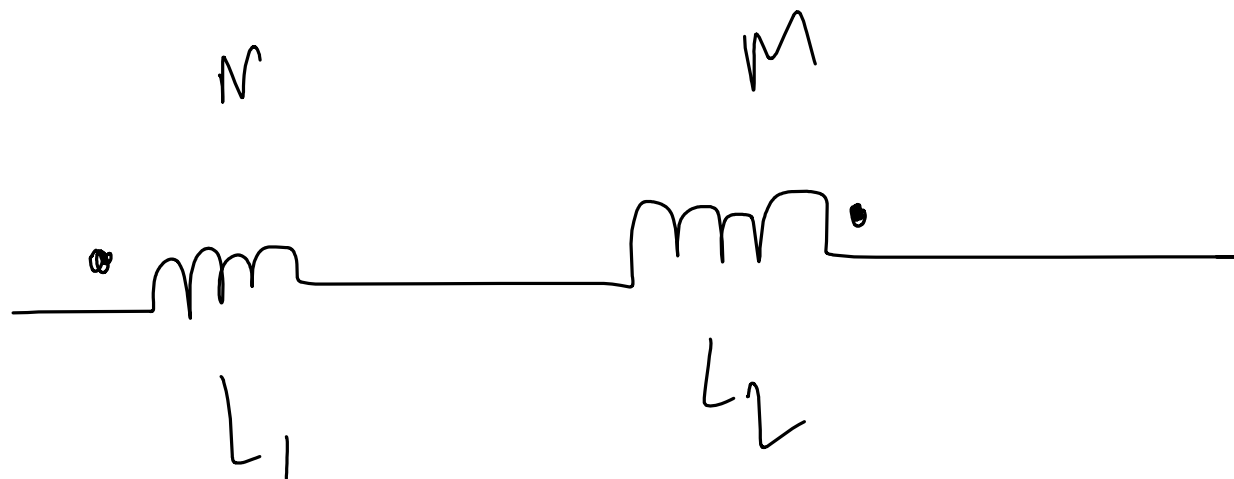


$$V_1 = -M \frac{di_2}{dt}$$

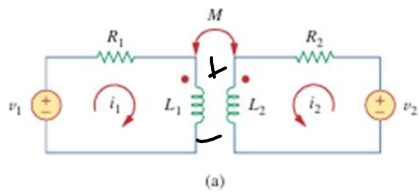




$$L = L_1 + L_2 + 2M$$



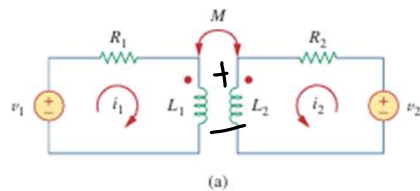
$$L = L_1 + L_2 - 2M$$



$$\text{Loop -1}$$

$$-V_1 + i_1 R_1 + L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} = 0$$

①



$j\omega$

Loop-2

$$-V_2 + i_2 R_2 + L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

= 0 (2)

eqⁿ - ① phasor form / freq. domain

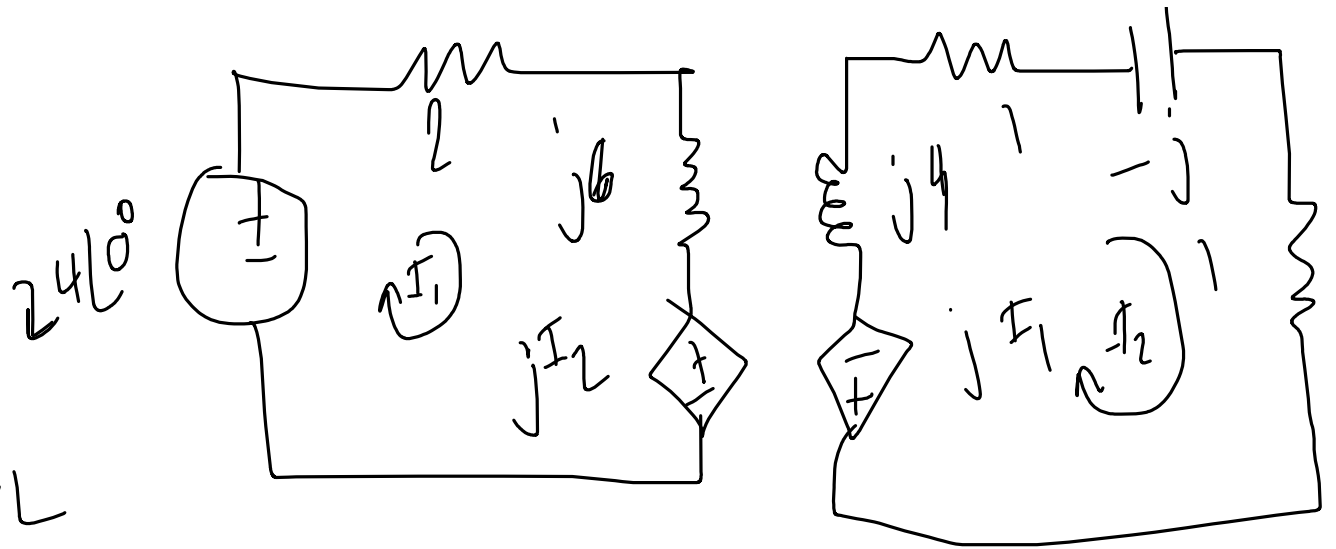
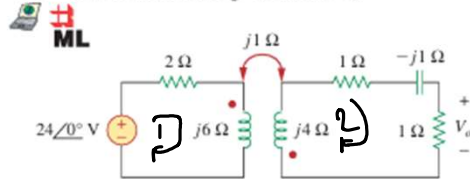
$$\hookrightarrow -V_1 + I_1 R_1 + \boxed{L_1 j\omega I_1} + M j\omega I_2 = 0$$

eqⁿ - ② $\Rightarrow$ freq. domain.

$$-V_2 + I_2 R_2 + L_2 j\omega I_2 + M j\omega I_1 = 0$$

FIG. 13.76

13.7 For the circuit in Fig. 13.76, find V_o .



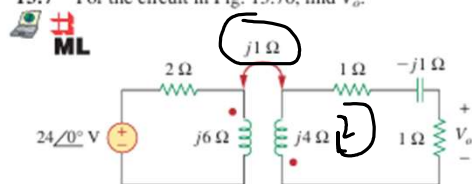
Loop-1 KVL

$$-24\angle 0^\circ + I_1 \times 2 + j6 I_1 + j I_2 = 0$$

$$\Rightarrow I_1 (2 + j6) + j I_2 = 24\angle 0^\circ \quad \text{--- (1)}$$

FIG. 13.76

13.7 For the circuit in Fig. 13.76, find V_o .



Loop-2

$$I_2 \times 1 - jI_2 + I_2 + j4I_2 + jI_1 \times I_1 = 0$$

$$\Rightarrow I_2 (1 - j + 1 + j4) I_2 + jI_1 = 0$$

$$\Rightarrow jI_1 + (2 + 3j)I_2 = 0 \quad \text{--- (1)}$$

① $\Rightarrow$ ②

$I_2, I_1 \rightarrow$ solve

$$V_0 = I_2 \times 1 \Omega$$